MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, LOW-POWER SCHOTTKY TTL, SCHMITT-TRIGGER POSITIVE-NAND GATES AND INVERTERS, MONOLITHIC SILICON

This amendment forms a part of MIL-M-38510/313A, dated 22 July 1983, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

TITLE: Delete and substitute as printed above.

1.2.3, outline letter C-2: Delete "X" and substitute "2".

1.2.3, add the following case outline:

"X C-2A (20-terminal, .350" x .350"), square chip carrier package".

1.3, thermal resistance, junction-to-case (θ_{JC}): delete and substitute the following:

"Thermal resistance, junction-to-case (θ_{JC}):
Cases A, B, C, D, X and 2 - - - - - - - - - - (See MIL-M-38510, appendix C)".

1.3, junction temperature (T_J), delete and substitute:

"Junction temperature (T_J) 2/ - - - - - - - +175°C.

Following footnote 1/ add footnote 2/ as follows:

"2/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions in accordance with method 5004 of MIL-STD-883."

PAGE 2

3.2.3, delete in its entirety and substitute the following:

"3.2.3 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in the specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers schematics shall be maintained by the agent activity and will be available upon request.

PAGE 4

3.6, second sentence: Delete in its entirety.

TABLE II: Delete "(preburn-in)".

The attached insertable replacement pages listed below are replacements for stipulated pages. When the new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

Replacement page	Page replaced
9	REPRINTED WITHOUT CHANGE
10	10

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TABLE II: Group A test requirements, class B devices, add "10, 11".

TABLE II: "Additional electrical subgroups for group C periodic inspections", delete in its entirety.

PAGE 5

4.2c Delete the first sentence and substitute the following:

"c. The percent defective allowable PDA shall be specified in MIL-M-38510."

4.4.3b, delete in its entirety.

PAGE 6

FIGURE 1, device type 01: delete "Case X" and substitute "Cases 2, X" make these same changes to device types 02 and 03.

PAGES 13 THROUGH 18

TABLE III, terminal designations, delete "Case 2" and substitute "Cases 2 and X".

PAGE 14

TABLE III, footnote 2, delete "Case X" and substitute "Case X, 2".

Make these same changes on pages 16 and 18.

PAGE 19

4.4.3c (3), delete and substitute:

"(3) Test duration: 1,000 hours, except as permitted by method 1005 of MIL-STD-883."

6.1: Delete in its entirety.

Custodian:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

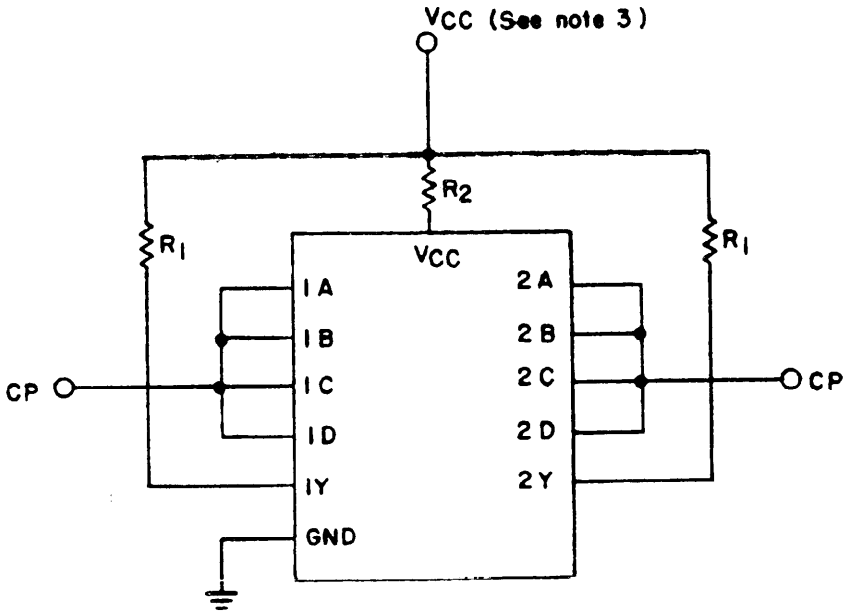
Preparing activity:
Air Force - 17

Agent:
DLA - ES

(Project 5962-1082)

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Device type 01



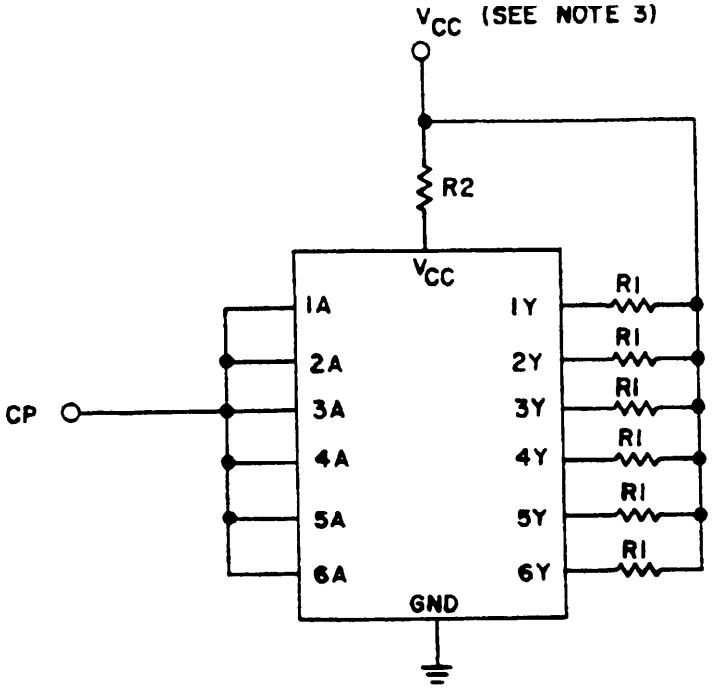
NOTES:
1. CP = 100 kHz ±50 percent square wave; duty cycle = 50 percent ±15 percent; V_{IL} = -0.5 V to 0.7 V; V_{IH} = 2.0 V to 5.5 V.
2. R_1 = 1 kΩ ±5 percent.
3. V_{CC} and R_2 shall be chosen to insure 5.0 V at device V_{CC} terminal.

FIGURE 3. Burn-in and life test circuits.

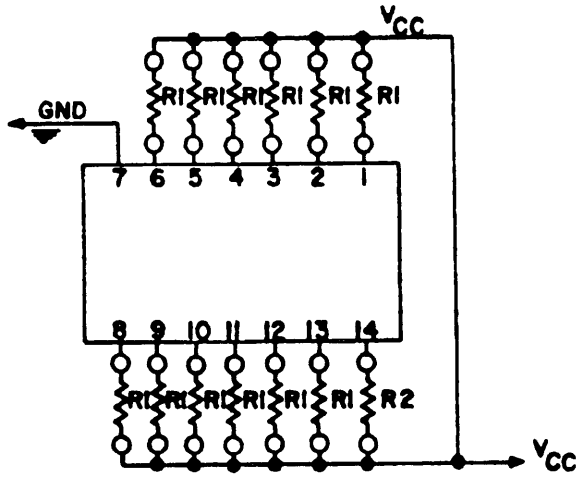
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Device type 02



NOTES:
1. CP = 100 kHz ±50 percent square wave; duty cycle=50 percent ±15 percent; V_{IL} = 0.5 V to 0.7 V; V_{IH} = 2.0 V to 5.5 V;
2. R1 = 1 kΩ ±5 percent.
3. V_{CC} and R2 shall be chosen to insure 5.0 V at device V_{CC} terminal.



NOTES:
1. High temperature (175°c), condition F, burn-in circuit only.
2. R1 = 2 kΩ.
3. V_{CC} shall be chosen such that 5.0 V minimum is present at device terminal.

FIGURE 3. Burn-in and life test circuits - Continued.

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